

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460361.8084 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1665 +/- 0.0088
Transit depth (Rp/Rs)^2	2.77 +/- 0.29 [%]
Orbital Inclination (inc)	83.89 +/- 1.39
Airmass coefficient 1 (a1)	1.07 +/- 0.034
Airmass coefficient 2 (a2)	-0.045 +/- 0.019
Scatter in the residuals of the lightcurve fit is	3.18 %
Best Comparison Star	#3 - [589, 384]
Optimal Aperture	2.71
Optimal Annulus	8.15
Transit Duration (day)	0.0819 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1665 ± 0.0088 vs 0.1368 (deviation 3.17σ)
Duration fit vs published	0.0819 d vs 0.0758 d (deviation 0.74σ)
Mid-time O–C	2.8 min
Depth SNR	17.8
Residual scatter	3.18 %

Light curve

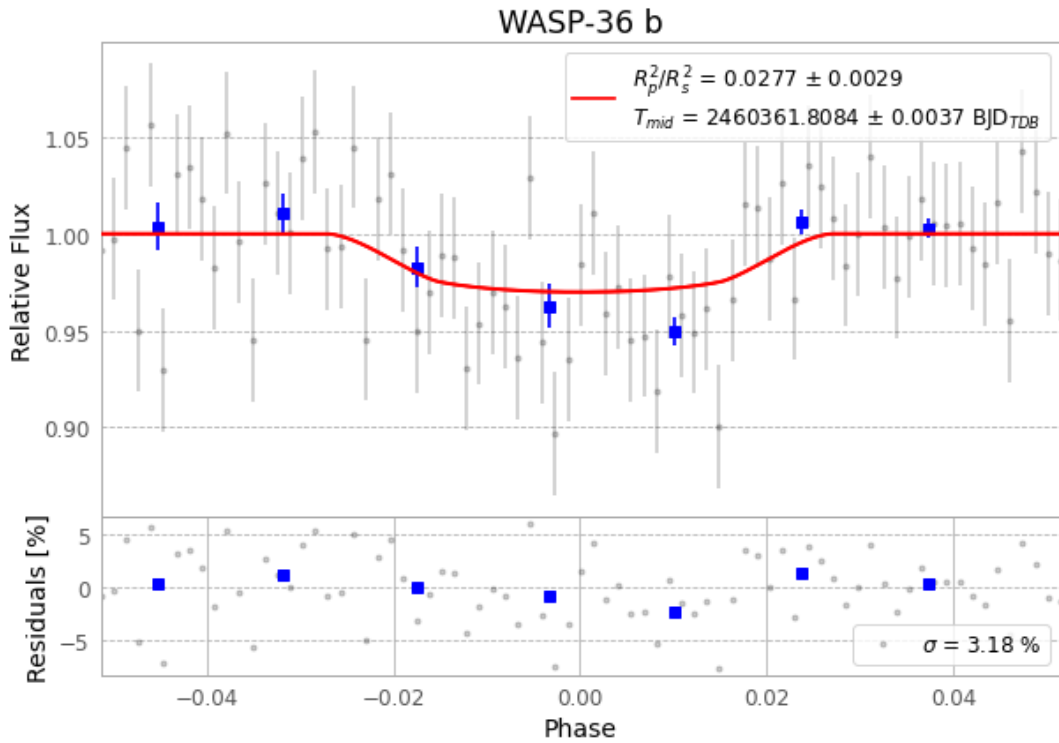


Figure 1 — FinalLightCurve_WASP-36 b_2024-02-20.png (SHA-256 7814746af325f80a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 214], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a328142310aa962e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-36240221052115.FITS → WASP-36240221090915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-240221.log (da9dbad0f9e2...), FinalLightCurve_WASP-36 b_2024-02-20.png (7814746af325...), FinalLightCurve_WASP-36 b_2024-02-20.csv (8354a33149f6...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 09:01Z.